

6.2.

$$V = \text{span} \left\{ \underbrace{(2, 1, 0, 1)^T}_{\vec{v}_1}, \underbrace{(1, 0, 1, 2)^T}_{\vec{v}_2} \right\} \subseteq \mathbb{R}^4$$

$$f(\vec{v}_1) = (4, 1, 2, 5)^T ; f(\vec{v}_2) = (2, 2, -2, -2)^T$$

$$(4, 1, 2, 5)^T = \vec{v}_1 + 2\vec{v}_2$$

$$(2, 2, -2, -2)^T = 2\vec{v}_1 - 2\vec{v}_2$$

ne jame B basis $B = (\vec{v}_1, \vec{v}_2)$;

$$\text{for } [f(\vec{v}_1)]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$[f(\vec{v}_2)]_B = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$\Rightarrow [f]_B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$\Rightarrow$ eigenvalues and eigenvectors

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = (1-\lambda)(-2-\lambda) - 4 = -2 - \lambda + 2\lambda + \lambda^2 - 4 =$$

$$= \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2) = 0$$

$$\lambda_1 = -3 ; \lambda_2 = 2$$

$$\left(\begin{array}{cc|c} 1-\lambda_1 & 2 & 0 \\ 2 & 2-\lambda_1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1-\lambda_1 & 2 & 0 \\ \lambda_1-1 & -\frac{1}{2}(1-\lambda_1)(2-\lambda_1) & 0 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{cc|c} 1-\lambda_1 & 2 & 0 \\ 0 & (1-\lambda_1)(2-\lambda_1)-4 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1-\lambda_1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$H_3 = \text{span} \left\{ \begin{pmatrix} 2 \\ \lambda_1-1 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

$\Rightarrow$ analogisch

$$H_2 = \text{span} \left\{ \begin{pmatrix} 2 \\ \lambda_2-1 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow [f^n]_{\mathcal{B}} = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}$$

$$\left[\underset{\substack{\parallel \\ \vec{u}}} {(5, 1, 3, 7)^T} \right]_{\mathcal{B}} = \underset{\substack{\parallel \\ \vec{w}}} {(1, 3)^T}$$

$$\Rightarrow \text{bestimme } C \text{ beliebig: } C = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & | & 1 \\ -2 & 1 & | & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & | & 1 \\ 0 & 5 & | & 5 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & | & 1 \\ 0 & 1 & | & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & | & -1 \\ 0 & 1 & | & 1 \end{pmatrix}$$

$$[\vec{w}]_c = (-1, 1)^T$$

$$[[f(\vec{w})]]_B = \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -(-3)^n \\ 2^n \end{pmatrix}$$

$$\Rightarrow [f^n(\vec{w})]_B = -(-3)^n \begin{pmatrix} 1 \\ -2 \end{pmatrix} + 2^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} (-1)^{n+1} 3^n + 2^{n+1} \\ 2 \cdot (-3)^n + 2^n \end{pmatrix}$$

$$\Rightarrow f^n(\vec{w}) = ((-1)^{n+1} 3^n + 2^{n+1}) \vec{v}_1 +$$

$$+ (2 \cdot (-3)^n + 2^n) \vec{v}_2$$

$$f^n(\vec{w}) = \left((-1)^{n+1} 3^n + 2^{n+1} \right) \vec{v}_1 + \\ + \left(2 \cdot (-3)^n + 2^n \right) \vec{v}_2,$$

$$\text{Ide } \vec{v}_1 = (2, 1, 0, 1) \quad ; \quad \vec{v}_2 = (1, 0, 1, 2)$$

$\Rightarrow$ per chiusura

$$= \begin{pmatrix} 5 \cdot 2^n \\ 2^{n+1} + (-1)^{n+1} \cdot 3^n \\ 2^n + 2(-3)^n \\ 4 \cdot 2^n + 3(-3)^n \end{pmatrix}$$
